

Mechatronic Skripsie Project

Biweekly Progress Report

for the project

Development of a High-Power Audio Amplifier with an Automated Thermal Management System

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Module / Project Code:	478
Lecturer / Supervisor:	Ms. Tessa Hall
Report Number:	03
Reporting Period:	31 March 2026 – 07 April 2026
Date Submitted:	08 April 2026

1 Overall Project Status

Overall Status:	At Risk
Current Phase:	Literature Review and Concept Generation/Selection
Report Summary:	This reporting period focused on researching amplifier designs and methods of cooling and clarifying what parameters will be measured during tests.

2 Objectives for the Reporting Period

The main objectives for this two-week period were:

1. Establish what circuits are in amplifier topology.
2. Find out what measureable parameters are important to success of amplifier.
3. Research thermal control methods.
4. Choose software simulation methods for testing and design.
5. Research Bridge Tied Load method for amplification.

3 Work Completed

3.1 Literature Review and Background Study

- Reviewed amplifier topology.
- Read Texas Instruments paper about bridge tied loads.
- Researched thermal control methods for heat management.

3.2 Circuit Design Progress

Circuit development completed:

- Designed constant current source.
- Designed current mirror.
- Designed input stage, with temporary VA stage.
- Considered feedback topology and compensation methods.

3.3 Thermal Control Progress

- Possible NTC thermistor and mosfet solution.
- Smallest ESP32 and 12V 4-pin PWM controlled fan.
- Temperature sensor already complete.
- Need to design fan driver circuit.

4 Work Planned for the Next Reporting Period

The planned work for the next two weeks is as follows:

1. Complete and test VA circuit in LTSpice.
2. Design cooling fan driver.
3. Complete calculations for current source and mirror in input stage.

5 Matters Requiring Lecturer / Supervisor Input

List any issues on which you would like feedback:

- Am I allowed to use the LM3886 IC?
- In a BTL/Parallel BTL configuration, where does the signal split? At what stage?
- What is linearity? Why is it good? Possibly in reference to the linear operation region of a transistor?
- What are the capabilities of the power supplies?
- Where could I obtain spice models for components?

Student Signature: _____

Date: _____